

Cherry Blossom Run

Two means

MATH 213

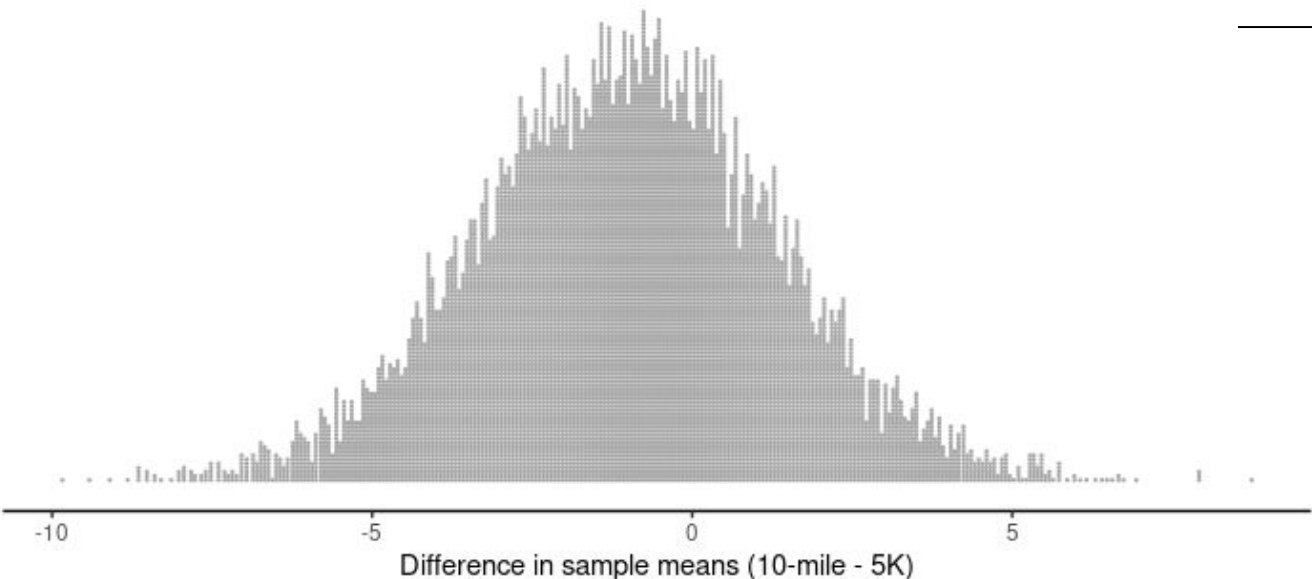
Group 1

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means
Mean age in 10-mile sample - mean age in 5K sample

quantile_025	Q1	median	Q3	quantile_975
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
-5.66	-2.6	-1	0.52	3.6405

mean	se
<dbl>	<dbl>
-1.023534	2.346699



What does each dot in the dotplot for the sampling distribution represent?

The difference in mean for a sample from the 10Mile runners and a sample from the 5K runners

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

-1, yes

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

Normal, yes. 95% of the data is within 2 standard deviations of the mean

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

1.513

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

1.796

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$\sqrt{(1.513^2 + 1.796^2)} = 2.348$

Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value?

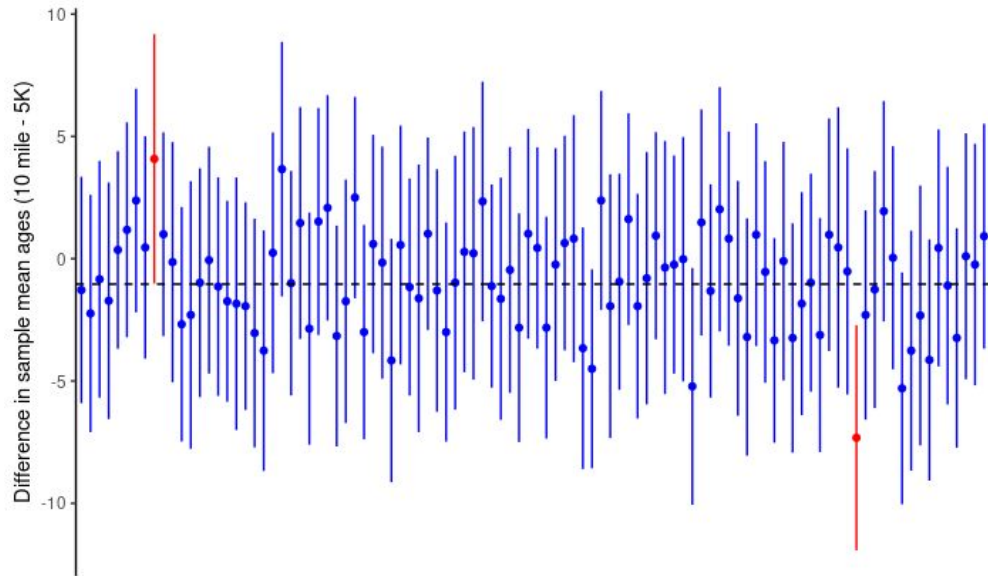
Somewhat close, the farthest is off by 0.25

replicate <int>	se_diff <dbl>
7983	2.325428
9754	2.135809
7244	2.284058
2719	2.100871
3996	2.375083
3531	2.312024
8423	2.307816
5936	2.306052
133	2.579468
6086	2.478994

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter? 95

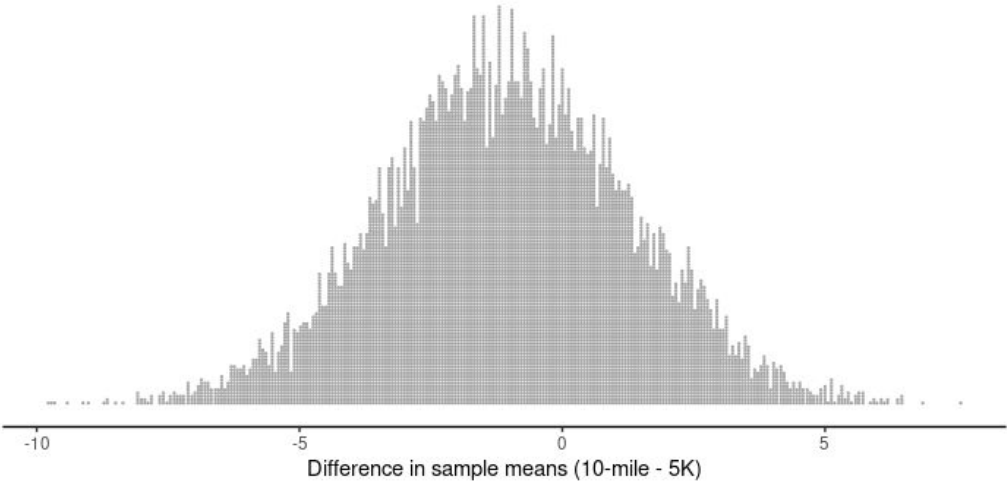
How many of your intervals actually contained the parameter? 98



Group 2

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means
Mean age in 10-mile sample - mean age in 5K sample



A tibble: 1 × 5

quantile_025	Q1	median	Q3	quantile_975
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
-5.74	-2.62	-1.08	0.54	3.52

1 row



A tibble: 1 × 2

mean	se
<dbl>	<dbl>
-1.061334	2.361384

1 row

What does each dot in the dotplot for the sampling distribution represent?

Each dot represents the difference in sample mean ages of the two cherry blossom run events.

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

Based on other sampling distributions, I would expect the center of this one to lie somewhere small, around 5 or so. The distribution ended up being centered at an even smaller number than I expected, being around -1.

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

I would expect this sampling distribution to be bell-shaped, which it is.

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

$$10.7/\text{sqrt}(50) = 1.513$$

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

$$12.7/\text{sqrt}(50) = 1.796$$

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$$SE_{\text{diff}} = \text{sqrt}(1.513^2 + 1.796^2) = 2.348$$

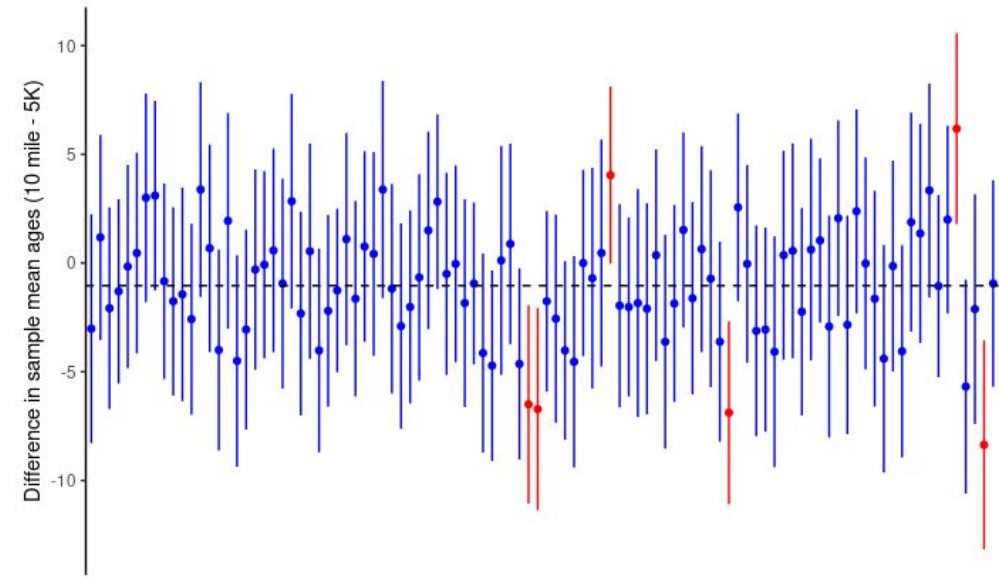
Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value?

replicate <int>	se_diff <dbl>
5157	2.407765
9416	2.537067
2833	2.445624
5482	2.091998
2444	2.148687
7678	2.521407
6221	2.290727
7507	2.388459
9286	2.105513
7314	2.186424

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?
I expect 95 of the 95% intervals to contain the population parameter.

How many of your intervals actually contained the parameter?
In the graph, 94 of the 95% intervals actually contained the parameter.

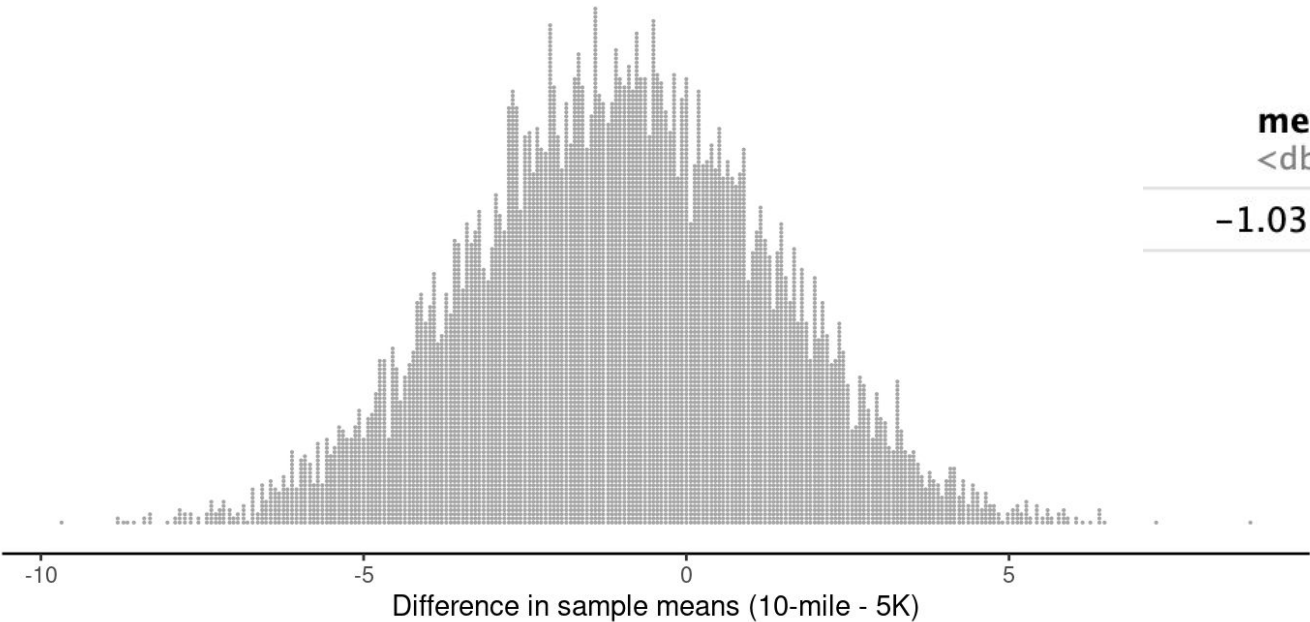


Group 3

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means
Mean age in 10-mile sample - mean age in 5K sample

quantile_025 <dbl>	Q1 <dbl>	median <dbl>	Q3 <dbl>	quantile_975 <dbl>
-5.64	-2.64	-1.04	0.58	3.52



mean <dbl>	se <dbl>
-1.0325	2.352405

What does each dot in the dotplot for the sampling distribution represent?

Each plot represents the result of one simulation, the difference in mean ages. Each plot is the difference for one pair of random samples.

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

I would expect the center to be close to the true difference in population means, so $37 - 38 = -1$ year. Looking at summary statistics we see the mean and median are -1.03 and -1.04 respectively.

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

I would expect it be be about normal, so it will be bell shaped.

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

$$10.7 / \text{square root } 50 = 1.51$$

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

$$12.7 / \text{square root } 50 = 1.50$$

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$$SE_{\text{diff}} = \text{square root } (1.51)^2 + (1.50)^2 = 2.35$$

Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value?

replicate <int>	se_diff <dbl>
8199	2.247422
6453	2.145182
5952	2.381481
3327	2.210662
9413	2.338716
3413	2.264469
7573	2.162716
3870	2.179444
6792	2.549622
4738	2.371692

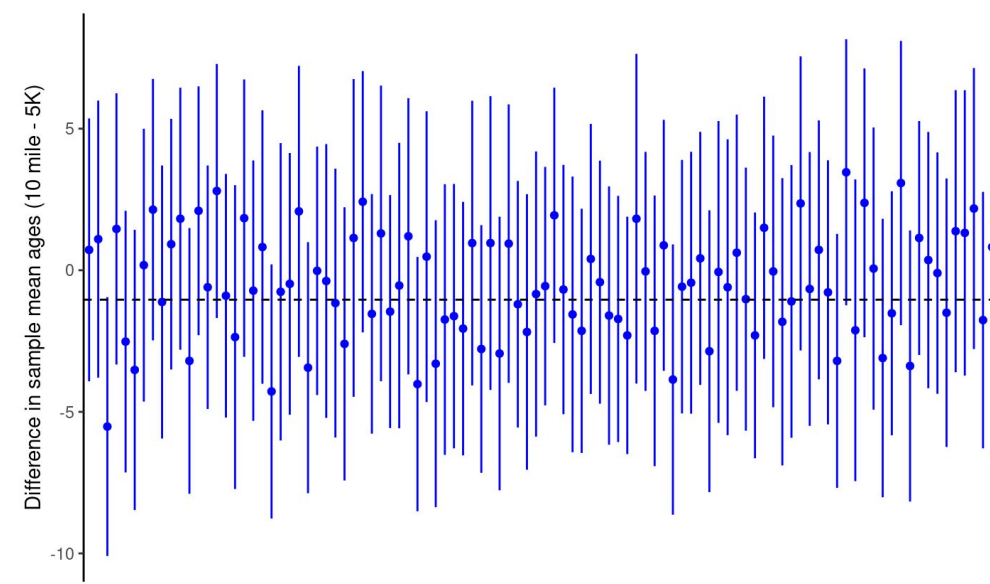
Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?

I expect to see 95 out of the 100

How many of your intervals actually contained the parameter?

100

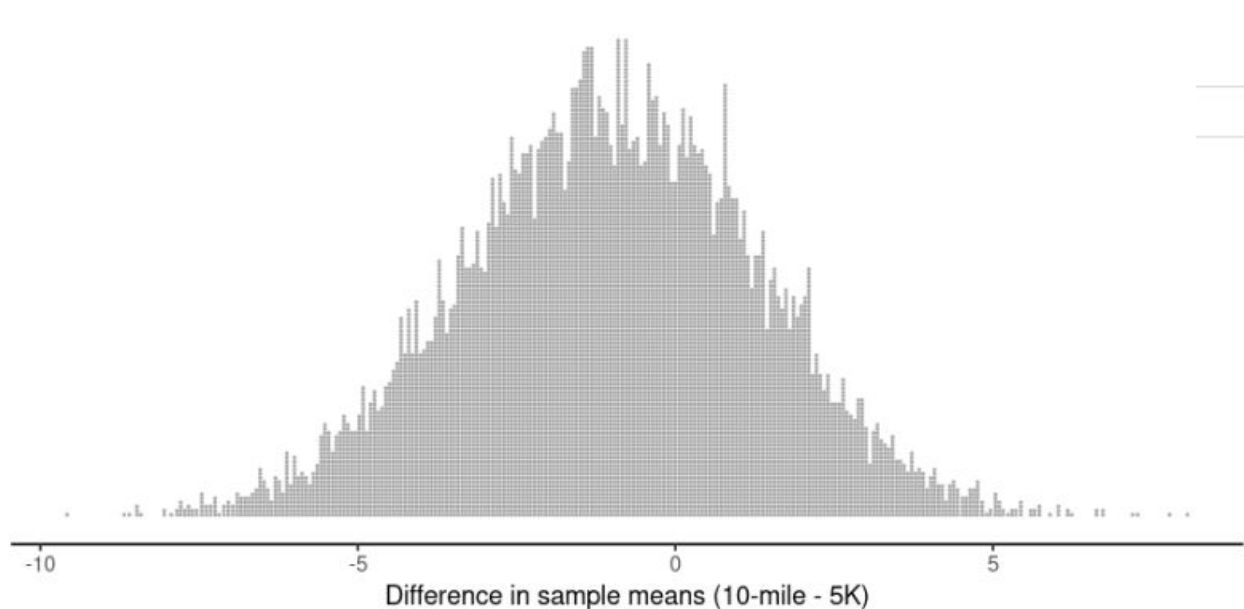


Group 4

Put your sampling distribution and the summary numbers for this distribution here.

quantile_025<dbl>	Q1<dbl>	median<dbl>	Q3<dbl>	quantile_975<dbl>
-5.58	-2.6	-1.02	0.58	3.48

Sampling distribution for the difference in means
Mean age in 10-mile sample - mean age in 5K sample



mean<dbl>	se<dbl>
-1.024362	2.333104

What does each dot in the dotplot for the sampling distribution represent?

Each dot is a difference in the mean age of a sample of 50 in the 10-mile group and a sample of 50 in the 5k group.

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

We expected the center to be at the difference in means of the real world mean ages for the 10mi-5k runners at -1. We were not disappointed to get a mean of -1.024.

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

We expected it to be a normal bell shape curve with the possibility of a slight right skew due to the older age of some of the runners. That is what we got although I do think the slight skew is smaller than i expected based off the significance of the skew in the real world data.

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

SE for 10 mile: $SD/\sqrt{n} = 10.7/\sqrt{50} = \mathbf{1.5}$

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

SE for 5K: $12.7/\sqrt{50} = \mathbf{1.8}$

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$E_{diff} = \sqrt{SE_{10}^2 + SE_{5k}^2} = \sqrt{1.5^2 + 1.8^2} = 2.34$

$-1 - 2 \times 2.34 = -5.68$

$-1 + 2 \times 2.34 = 3.68$

95% confidence interval: (-5.68, 3.68)

Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value?

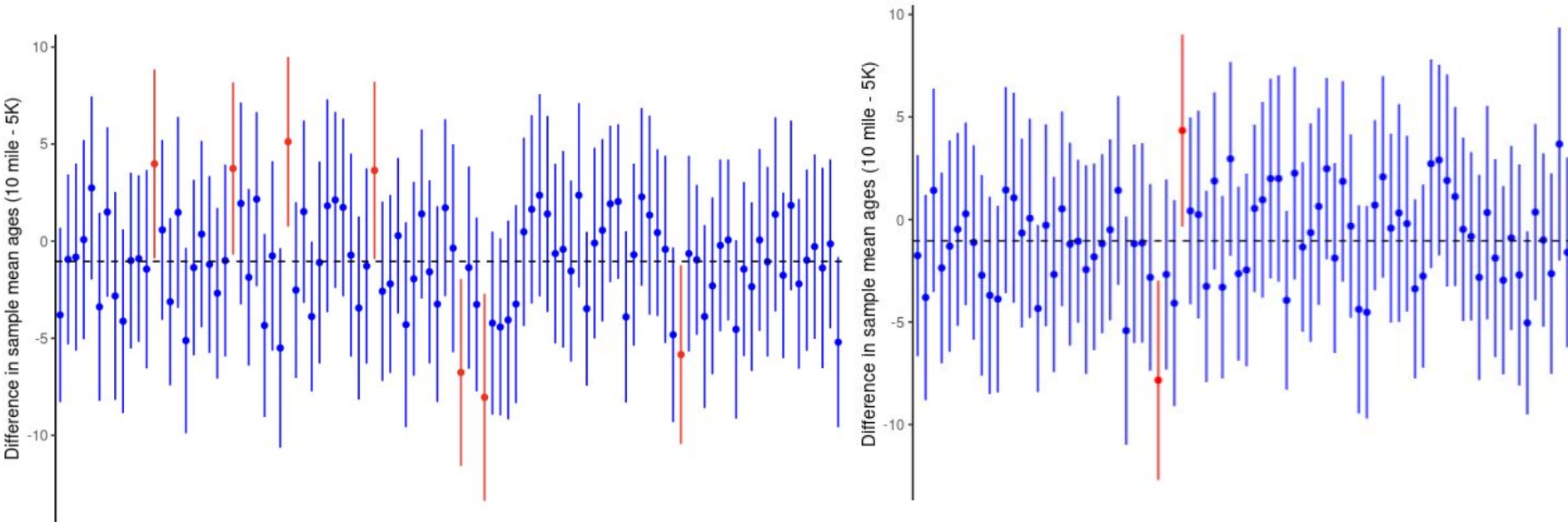
replicate <int>	se_diff <dbl>
9529	2.364416
6840	2.433798
5569	2.231493
8231	2.349734
8059	2.246906
6433	2.147051
2361	2.035229
3944	2.245447
9541	2.087328
8438	2.271544

The 10 estimated SEs all are within about .33 years of the actual value of 2.35 years.

Put your graph testing the new confidence-interval procedure here.

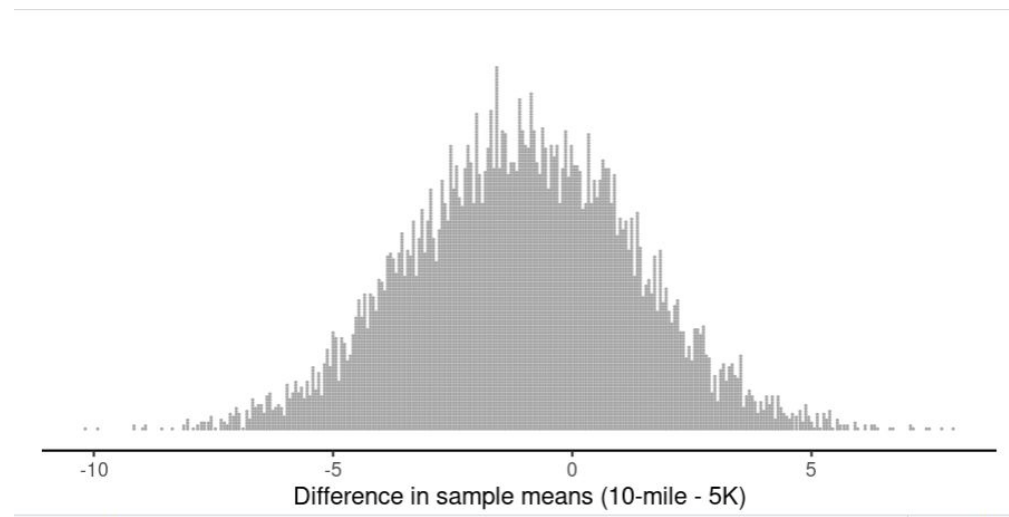
How many of the 100 95% intervals do you expect to contain the population parameter? **About 95**

How many of your intervals actually contained the parameter? **93 / 98 (ha!)**



Group 5

Put your sampling distribution and the summary numbers for this distribution here.



Summary Statistics

quantile_025<dbl>	Q1<dbl>	median<dbl>	Q3<dbl>	quantile_975<dbl>
-5.72	-2.66	-1.04	0.58	3.54

What does each dot in the dotplot for the sampling distribution represent?

Each dot in the dotplot represents a difference in means between a 10k and 5 mile racers mean ages

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

I expected it to be zero, but the real one is -1.04

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

It has a bell shaped curve which is what i expected

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?
1.5

What is the standard error for the sample mean age in samples of size 50 from the 5K group?
1.8

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$$\text{sqrt}(1.5^2+1.8^2)=2.43$$

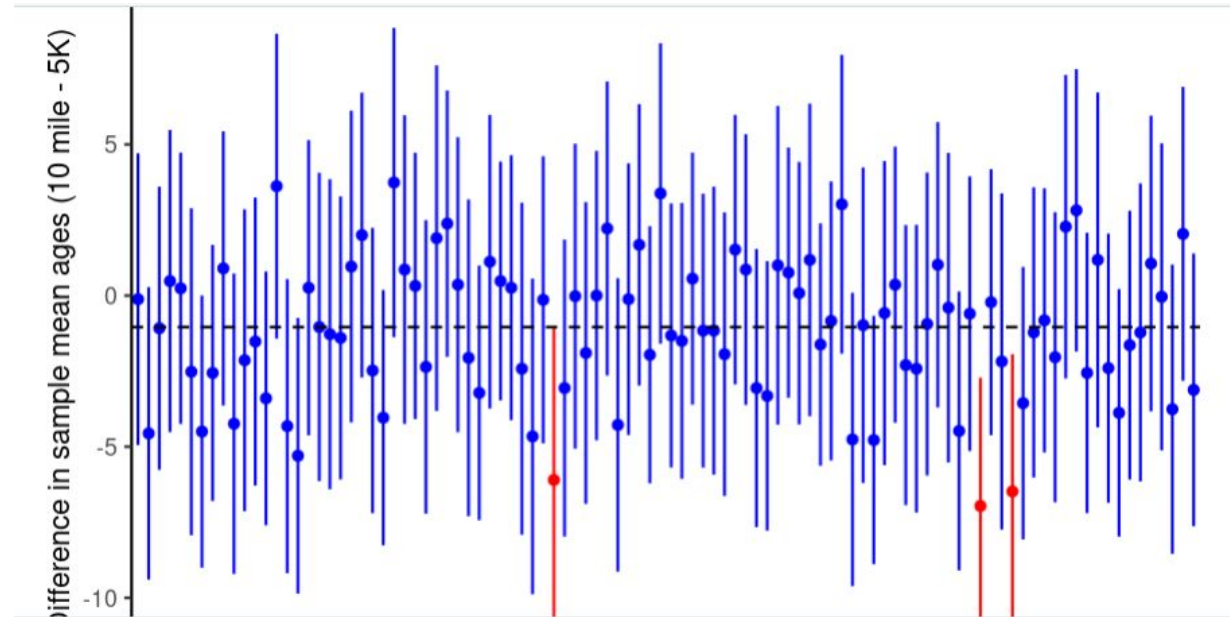
Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value?

One estimate, 8050 had a difference in se which was 2.36

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter? Above 90

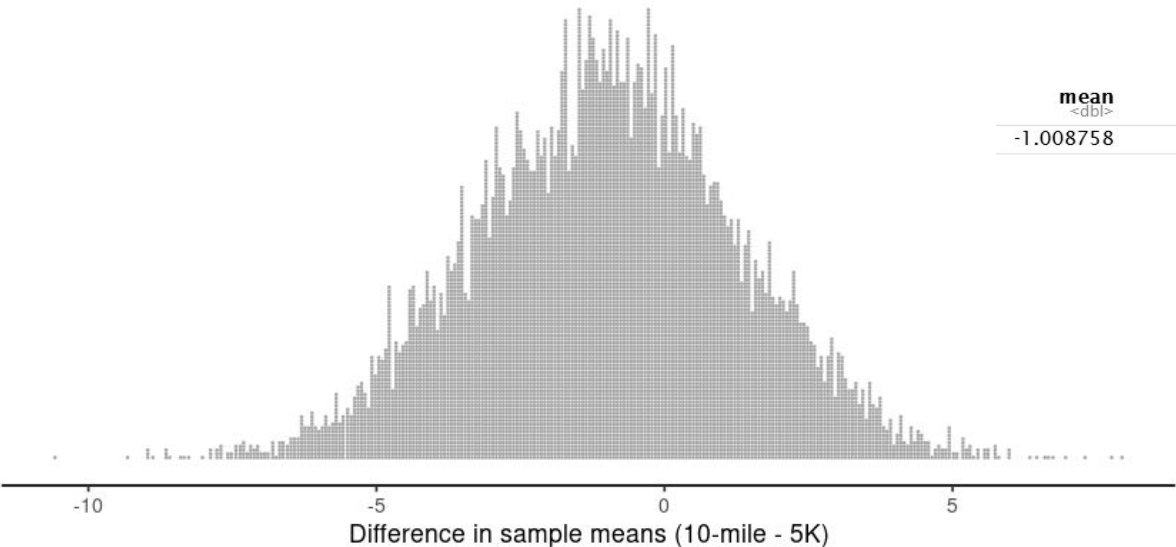
How many of your intervals actually contained the parameter? 97



Group 6

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means
Mean age in 10-mile sample - mean age in 5K sample



mean	se
<dbl>	<dbl>
-1.008758	2.303433

quantile_025	Q1	median	Q3	quantile_975
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
-5.48	-2.56	-0.98	0.52	3.44

What does each dot in the dotplot for the sampling distribution represent?

Difference in means between 10-mile and 5K random samples of size 50

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

The center should be at -1; according to the mean/median, yes, this is where we expect it to be

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

Bell shape; yes, it has this shape. We would check using a 95% confidence interval within 2 std. dev. of the population parameter?

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

1.51

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

1.80

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$\text{sqrt}((1.51)^2 + (1.80)^2)$

= 2.35

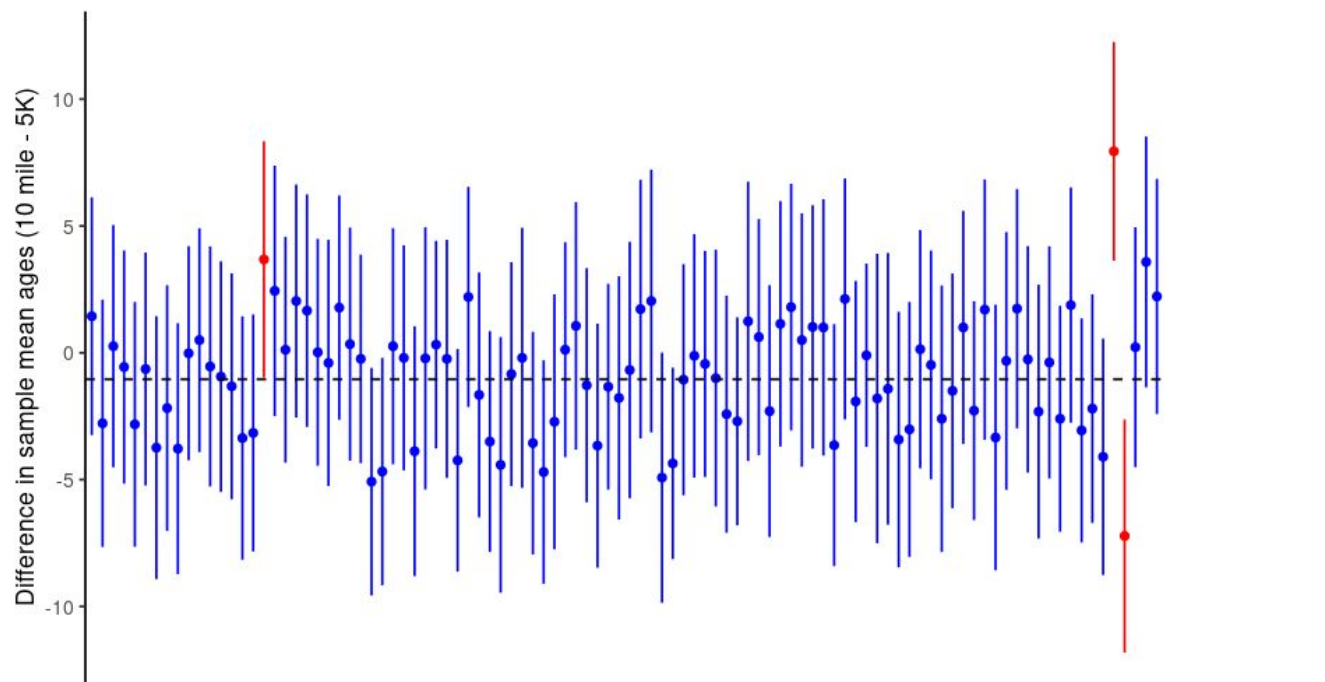
Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value? *It's pretty close, the most different estimation is 0.3 off from the actual standard error.*

se_diff <dbl>
2.136753
2.226464
2.446956
2.303148
2.393327
2.400451
2.415699
2.146310
2.438618
2.617567

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter? 95

How many of your intervals actually contained the parameter? 97

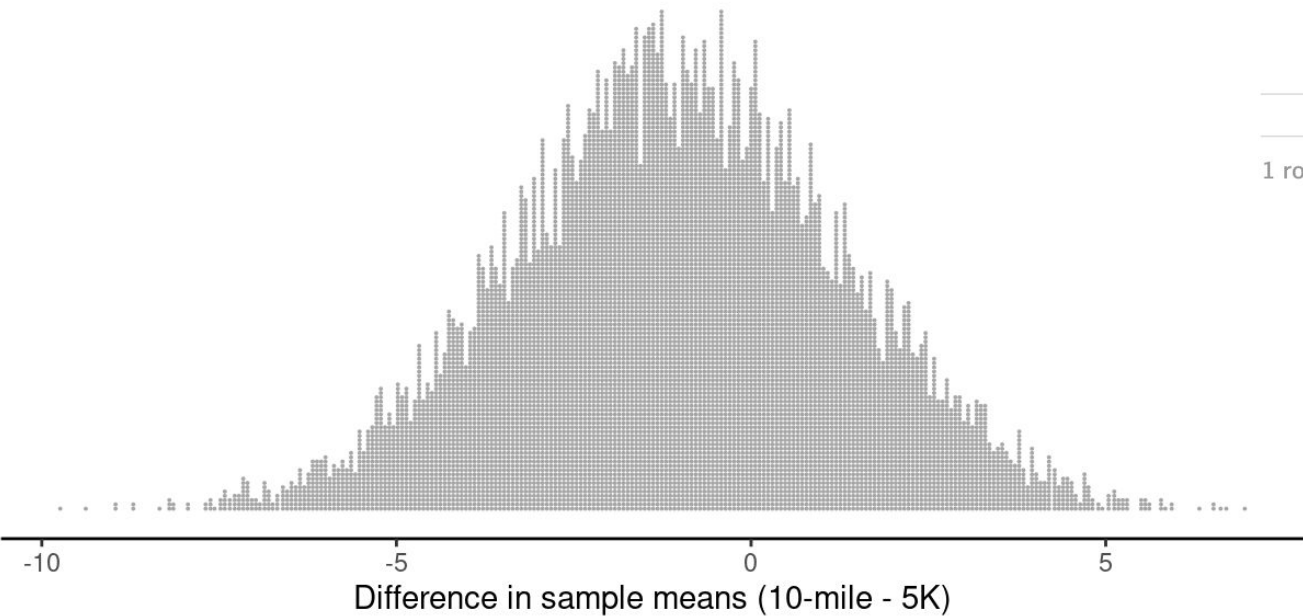


Group 7

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means

Mean age in 10-mile sample - mean age in 5K sample



mean <dbl>	se <dbl>
-1.04695	2.320581

1 row

What does each dot in the dotplot for the sampling distribution represent?

Difference in means from a sample of 500000

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

At the true difference of means. It is.

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

Normal bell curve. It does, I can check by ?

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

2.320581

Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value? Pretty close mostly

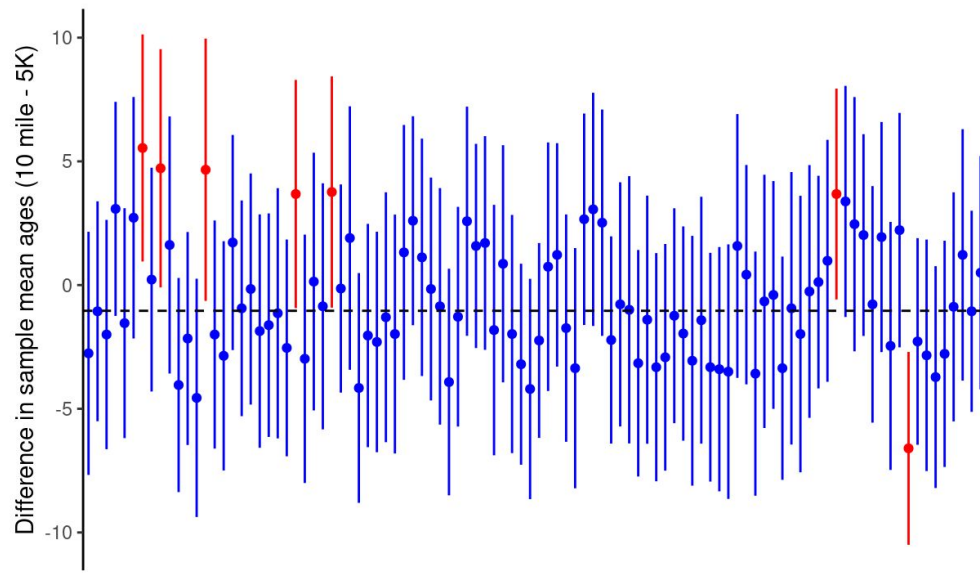
replicate <int>	se_diff <dbl>
3481	2.437794
9950	2.319418
2525	2.321971
7072	2.173016
1600	2.502629
9915	2.222999
3790	2.303547
7847	2.443966
3995	2.186095
7423	2.667758

1-10 of 10 rows

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter? 95

How many of your intervals actually contained the parameter? 93



Group 8

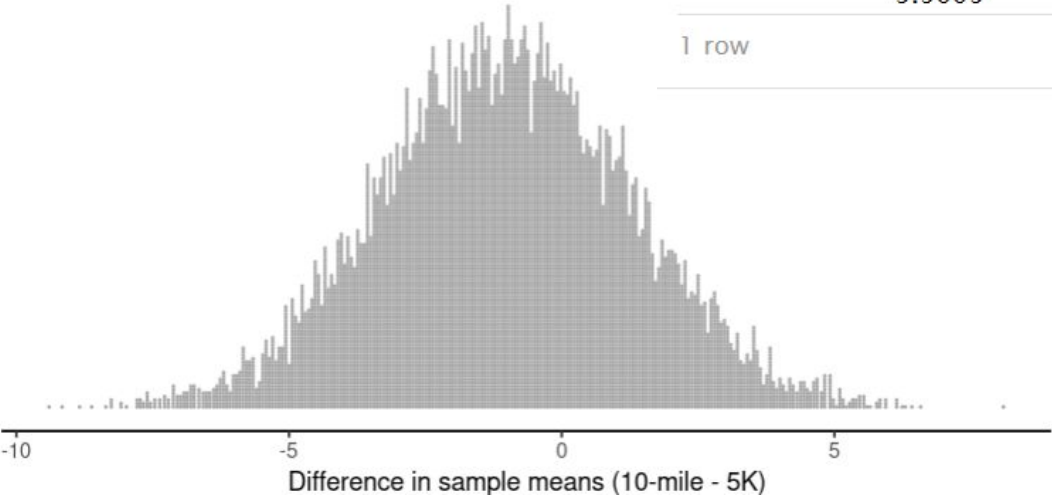
Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means

Mean age in 10-mile sample - mean age in 5K sample

quantile_025<dbl>	Q1<dbl>	median<dbl>	Q3<dbl>	quantile_975<dbl>
-5.5005	-2.585	-1.04	0.52	3.5

1 row



What does each dot in the dotplot for the sampling distribution represent?

The difference in mean age for a sample.

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

At the true difference of means.

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

Bell curve

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

1.513

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

1.796

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$$\text{sqrt}((1.513)^2 + (1.796)^2) = 2.348$$

$$-1 - 2(2.348) = -5.696$$

$$-1 + 2(2.348) = 3.696$$

Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value?

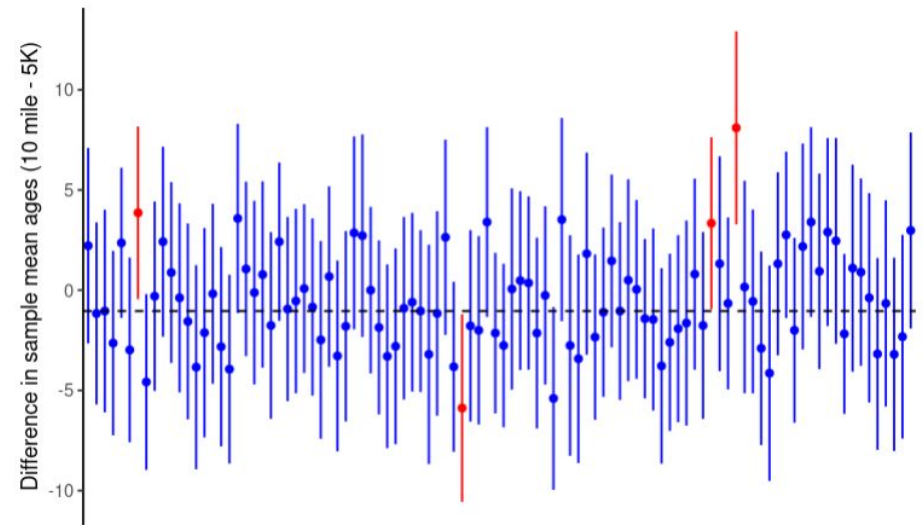
replicate <int>	se_diff <dbl>
5678	2.283436
7705	2.070891
2276	2.379333
763	2.210596
4339	2.184939
8025	2.177850
7409	2.340237
3840	2.550341
5783	2.356469
8461	2.577366

Pretty close

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?
95

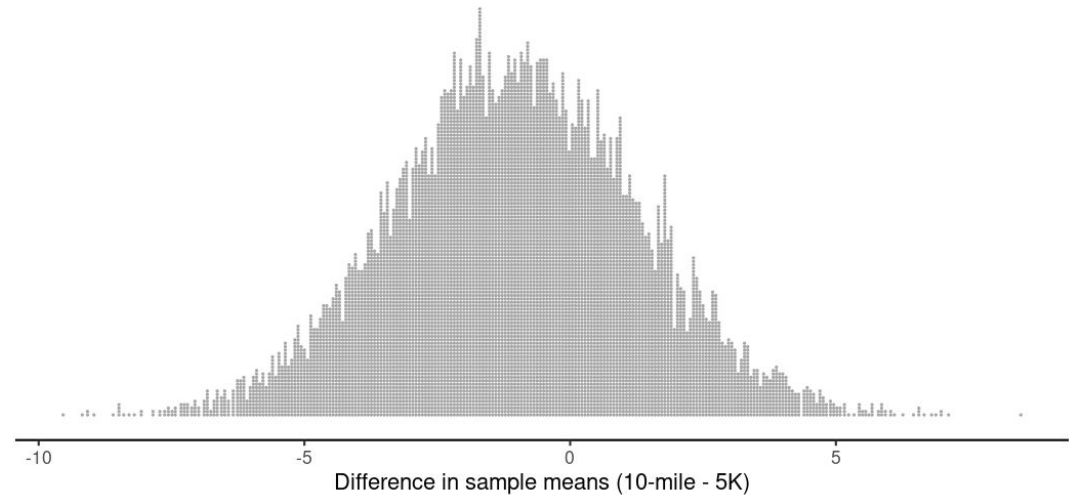
How many of your intervals actually contained the parameter?
96



Group 9

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means
Mean age in 10-mile sample - mean age in 5K sample



quantile_025<dbl>	Q1<dbl>	median<dbl>	Q3<dbl>	quantile_975<dbl>
-5.52	-2.58	-1.04	0.54	3.58

What does each dot in the dotplot for the sampling distribution represent?

Each dot represents the difference in the mean between the 10-mile group and the 5k group

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

We would expect the center to be at -1. The center ended up being where we expected as R calculated it to be at -1.02

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

We would expect it to be a normal bell shaped curve, which it ended up being.

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

$$10.7 / \sqrt{50} = 1.5$$

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

$$12.7 / \sqrt{50} = 1.8$$

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$$\sqrt{1.5^2 + 1.8^2} = 2.34$$

$$-1 \pm 2 * 2.34$$

$$-1 + 2 * 2.34 = -5.68$$

$$-1 - 2 * 2.34 = 3.68$$

95% confidence interval: (-5.68, 3.68)

Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value?

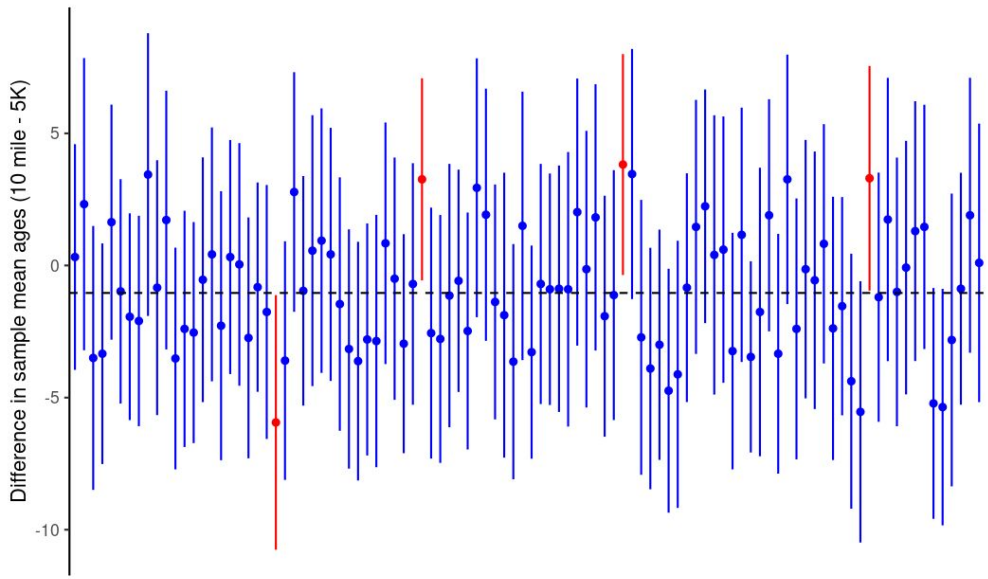
The 10 estimated standard errors are pretty close

replicate <int>	se_diff <dbl>
2891	2.380989
6477	2.294848
2263	2.165963
1985	2.553544
6333	2.360949
988	2.315627
9720	2.173437
6961	2.333200
2061	2.131857
5673	2.277795

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?
I would expect 95 of the 95% intervals to contain the population parameter
How many of your intervals actually contained the parameter?

96

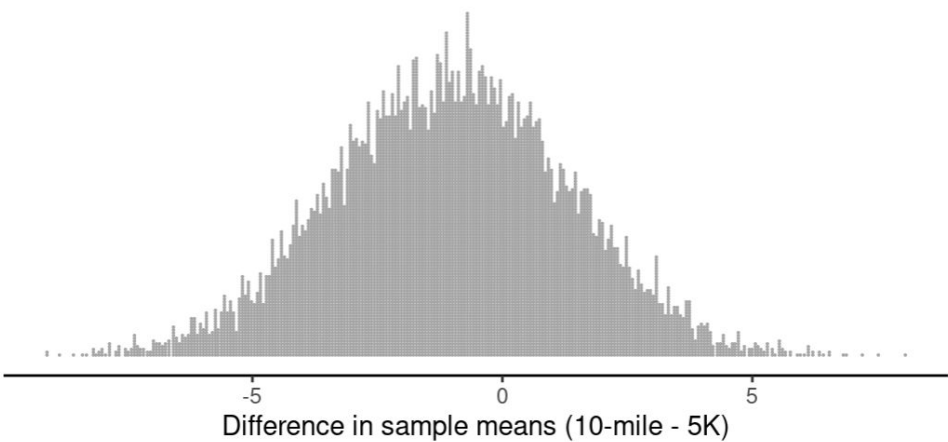


Group 10

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means

M Difference in Two Means sample - mean age in 5K sample



A tibble: 1 × 5

quantile_025	Q1	median	Q3	quantile_975
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
-5.68	-2.66	-1.02	0.56	3.56

1 row

A tibble: 1 × 2

mean	se
<dbl>	<dbl>
-1.035752	2.360927

1 row

What does each dot in the dotplot for the sampling distribution represent?

Each dot on the dotplot shows a random sample of 50 people in each race 10k or 5k.

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

The center of the graph should be around -1. The graph is just to the left of 0 so it is on -1, as well as the median is showing us that the center of the graph is -1.

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

I would expect the shape of the graph to be bell shaped. It is very bell shaped.

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?

The standard error for sample mean of 10 mile group is 1.513

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

The standard error for a sample of the 5k group is 1.796

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

The standard error for the difference in sample means is going to be 2.348.

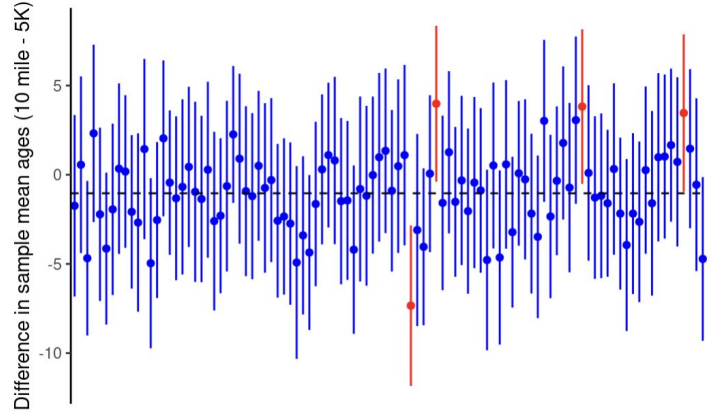
Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value? They are pretty close because the true value is 2.438 and these 10 numbers are all either just below it or just greater than it.

replicate <int>	se_diff <dbl>
6342	2.569531
7802	2.059578
6012	2.545799
3820	2.533435
2228	2.364489
6761	2.310335
9427	2.331423
3903	2.437630
8256	2.486946
3671	2.344844

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?
Out of the 100 intervals there is only 4 that is outside of the 95% confidence interval, there is 96 that are in the confidence interval.

How many of your intervals actually contained the parameter?
96 of the intervals contain the population parameter

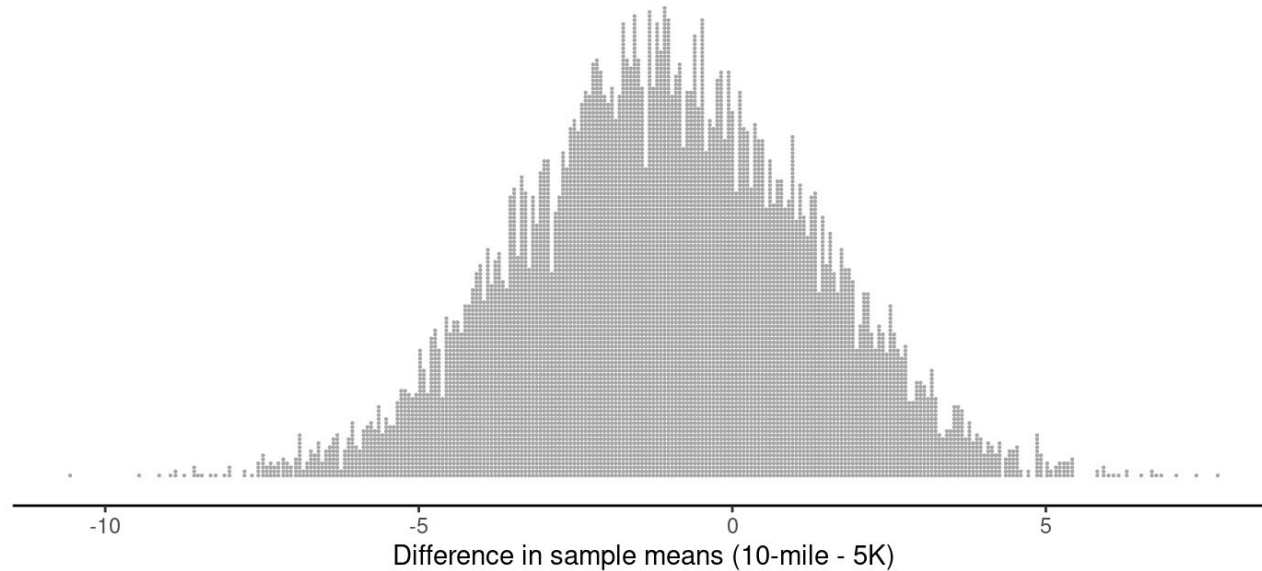


Group 11

Put your sampling distribution and the summary numbers for this distribution here.

Sampling distribution for the difference in means

Mean age in 10-mile sample - mean age in 5K sample



What does each dot in the dotplot for the sampling distribution represent?

Each dot in the sampling distribution represents the difference in means between random samples of the two events

Judging from your previous experience with sampling distributions, where would you expect the center of this sampling distribution to be? Is it where you would expect?

Centered around -1, which is true

Judging from your previous experience with sampling distributions, what shape would you expect this sampling distribution to have? Does it have this shape? How would check?

normal/bell shaped, which is true. We can verify this by checking if 95% of the data is within 2SE of the mean

Try out the formula for the standard error for a difference in means:

What is the standard error for the sample mean age in samples of size 50 from the 10-mile group?
The standard error for sample mean of 10 mile group is

$$SE = 10.7/\sqrt{50} = 1.51320851174$$

What is the standard error for the sample mean age in samples of size 50 from the 5K group?

$$SE = 12.7/\sqrt{50} = 1.79605122421$$

What is the standard error for the difference in sample mean ages for pairs of samples of size 50 from the 10-mile group and the 5K group? (Use the “Pythagorean Theorem,” showing some of your work.)

$$SE_{diff} = \sqrt{SE_{10m}^2 + SE_{5k}^2}$$

$$SE_{diff} = \sqrt{1.513^2 + 1.796^2}$$

$$SE_{diff} = 2.348$$

Put your table to 10 estimated standard errors here. The actual standard error for the difference in sample means is around 2.35 years. How close are the 10 estimated standard errors to this true value? → the 10 estimated values are very close give or take about 0.33 SE_diff in either direction.

replicate <int>	se_diff <dbl>
119	2.417835
8228	2.473345
3412	2.495040
917	2.193184
7111	2.261760
375	2.205903
7017	2.259494
650	2.448285
5457	2.038695
1721	2.416109

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?

How many of your intervals actually contained the parameter?

